

Long bristles deter insect-eating birds and parasitic wasps

Tiger moth caterpillar



Fake fur

Insects and spiders often have furlike bristles on parts of their bodies. Instead of being made of keratin, these hairs are made of chitin—the substance that forms insect exoskeletons. Large moths and bumblebees have coats of bristles that keep their flight muscles warm, but caterpillars use theirs mainly for self-defense. Caterpillar bristles may look harmless, but when touched, they break easily and release poisonous chemicals that help to ward off predators.

Furry wet suit

When an otter dives, its outer fur gets wet, but its skin stays completely dry. This happens because, like many land mammals, the otter has a double coat. This Asian small-clawed otter's long guard hairs form the coat's surface. They point from the head toward the tail, creating a streamlined shape for swimming. The otter's underfur hairs are finer and much more densely packed. They are also kinked, which helps to trap a layer of air. This air keeps water out and ensures that the otter's body stays warm and dry.



Camouflage colors

Crouching in dry grass, a female leopard is camouflaged from prey by its beautifully patterned coat. Spotted markings like these are created mainly by melanin—a chemical pigment stored in individual hairs. Melanin can produce a wide range of hues, from pale yellow to brown and black. This pigment also occurs in animal skin and in many other body coverings, such as feathers and scales.



Stripes of dark and light colors blend in with woodland surroundings



Arctic protection

Polar bears can survive the bitter cold on the Arctic's drifting ice and in the sea. They have a thick fur coat and an insulating layer of fat, called blubber, which protects against the freezing temperatures.

Hair's hollow center traps air



Hollow hairs

This microscopic view of polar bear fur shows that its guard hairs are hollow and filled with air. These hairs work like an insulating jacket and greatly reduce heat loss, helping to keep the bear's internal body temperature at 98.6°F (37°C).



Heat portrait

This infrared photograph reveals the surface temperature of a polar bear's body. The white and yellow areas are coldest, proving that the bear has good body insulation. Exposed body parts, such as the nose, look red and show where the bear is losing the most heat.